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Mother–Infant Attachment in the Squirrel Monkey: Adrenal Response to Separation[1]

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The pituitary–adrenal response following separation was evaluated in mother and infant squirrel monkeys. Four mother–infant pairs and a pregnant female, living in a social group, were the subjects of this experiment. The plasma cortisol levels of the mothers and infants were determined after the following conditions: (1) basal levels at 1100 hr, (2) 30 min after momentary separation and reunion, (3) 30 min after infant removal from the group, and (4) 30 min after mother removal from the group. The levels of plasma cortisol were significantly elevated in both mothers and infants following separation and the response was not reduced by the presence of familiar animals (e.g., separated infants which were "aunted" by the pregnant female). Separation followed by immediate reunion did not result in elevated values. These data indicate that a specific attachment relationship develops between mother and infant, and that the agitation following separation is reduced only by reunion with the object of attachment.

Mother–infant attachment in simian primates has been measured by the occurrence of affiliative and rejection behaviors, the infant's recognition and preference for the maternal figure and the behavioral responses to involuntary separation (Hansen, 1966; Hinde and Spencer-Booth, 1971; Kaufman and Rosenblum, 1967; Rosenblum, 1971). The high levels of affiliative behavior which occur immediately postpartum function to ensure the infant's survival as well as to initiate the attachment process. Initially, the infant's behaviors (e.g., clinging and suckling) are not directed specifically toward the genetic mother or even the nurturant one

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but rather are responses to objects with certain stimulus characteristics (Harlow and Zimmerman, 1959). The expression of these behaviors appears to be self-reinforcing, tending to reduce the infant's arousal, and if directed toward the same object for long duration, the object will become the specific focus of emotional attachment (Mason, 1971). Research on the infant's ability to distinguish the maternal figure indicates that a specific attachment develops rapidly although the ontogeny of the infant's perceptual ability may vary between species and sensory modalities (Kaplan and Schusterman, 1972; Rosenblum and Alpert, 1974).

Once the attachment bond is established, the presence of the object provides emotional security and its absence evokes the infant's agitation. The ability to cling to a "caregiver figure," for example, increased the threshold for painful electrical stimulation in chimpanzee infants (Mason and Berkson, 1962); and the presence of an attachment object reduced the stressful reaction of young macaques to an unfamiliar environment (Hill *et al.*, 1973). Moreover, despite the rapid development of independence in infant monkeys, involuntary separation from the mother during the first year has repeatedly been shown to result in marked disturbance behaviors (Hinde and Spencer-Booth, 1971; Jones and Clark, 1973; Kaplan, 1970; Kaufman and Rosenblum, 1967; Preston *et al.*, 1970; Rosenblum and Youngstein, 1974; Seay and Harlow, 1965). Assessment of the response to separation has also been used in studies on the nature of attachment in human children (Bowlby, 1973). The intensity of the response is typically accentuated in unfamiliar environments or by the presence of a stranger and may be ameliorated by the presence of a familiar person (Kagan, 1974; Sroufe *et al.*, 1974). Similarly, in those studies on monkeys in which nonmother animals engaged in compensatory behaviors toward the separated infants, the behavioral disturbance was reduced or absent (Kaufman and Rosenblum, 1967; Preston *et al.*, 1970; Rosenblum, 1971).

In the squirrel monkey, the subject of the following experiment, studies on mother–infant separation have revealed similar findings on the behavioral consequences of separation (Jones and Clark, 1973; Kaplan, 1970). Prior research in our laboratory has demonstrated also that both mother and infant squirrel monkeys undergo significant elevations in plasma cortisol levels following a 30-min separation (Mendoza *et al.*, in press). The dyads in that study were individually housed and therefore were alone when separated. The present experiment was conducted on group-living monkeys in order to assess the effect of the presence of familiar animals and environment on the separated mother and infant. Rosenblum (1971) has reported that infants which are "aunted" by nonmother females emit few distress vocalizations during sustained separation. However, it is possible that the separated infants are still highly aroused without exhibiting behavioral indications of disturbance. Since numerous studies have demonstrated that plasma levels of corticosteroids

are a sensitive measure for assessing changes in arousal and subtle shifts in the mother–infant relationship (e.g., Levin *et al.,* 1976; Smotherman *et al.,* 1976), we have utilized the responses of the pituitary–adrenal system to evaluate further the nature of attachment in the squirrel monkey.

METHOD

Subjects

Four mother–infant pairs and an adult female squirrel monkey were the subjects of this experiment. The five females were imported as adults from Guyana and were well acclimated to laboratory conditions. The nonmother female was term pregnant, a period of high maternal responsiveness (Rosenblum, 1972), and served as a potential aunt for the infants following separation in the group. Aunting is a common species-specific behavior pattern in that infants are often carried dorsally by young nonlactating females or by females familiar with the infant's mother. The infants (three females and one male) were lab born and were 8–10 weeks of age at the beginning of the experiment.

The monkeys were housed as a group in an outdoor enclosure (4.25 m^3) constructed with wire-mesh walls and containing three sitting perches. They were exposed to natural climatic conditions although heat lamps provided supplemental warmth. Food and water were available *ad lib.*

Procedure

Blood samples for cortisol determination were collected from the mother–infant dyads across a 5-week period between 1100 and 1200 hr. Blood samples from each pair were obtained a minimum of a week apart and only one pair was sampled per day. The order of the experimental conditions was balanced across pairs. The four conditions were: (1) Basal, the dyad was caught immediately and blood from both mother and infant was sampled; (2) Separation Reunion, the dyad was caught and separated momentarily, and blood samples were collected 30 min later; (3) Infant Removal, the specified infant was removed from the group and housed in a different cage, and blood from both mother and infant was sampled after 30 min; and (4) Mother Removal, the specified mother was removed and housed alone, and blood samples from both mother and infant were obtained after 30 min. During the Mother Removal condition the group was observed discretely via a small "peephole" at 10-min intervals in order to determine whether the separated infant was aunted. At each interval the observer recorded whether the separated infant was being carried by an adult female. It should be noted that the Removal conditions confound the effects of separation and novel environment for the removed animal; the data on the transferred subjects are included only for the purpose of comparison with the responses of separated animals left within the group.

The blood-sampling procedure involved rapid capture of the dyad, anesthetization with ethyl ether, and collection of 0.8 ml of blood in a heparinized syringe via cardiac puncture. Blood samples were obtained within 3 min following experimenter entrance into the enclosure, thereby minimizing hormonal elevation due to capture. The blood was centrifuged at 2000 rpm and plasma was extracted and fast frozen for subsequent determination of cortisol levels by radioimmunoassay. Plasma cortisol was assayed using antibody F21-53 from Endocrine Sciences, Tarzana, California as described by Klemm and Gupta (1975). Prior to assay the plasma was extracted with ethyl acetate, and the assay was performed according to the Endocrine Sciences method except that the heat step and methanol were omitted (for further details, see Mendoza *et al.*, in press).

The levels of plasma cortisol determined for each subject as well as the median value for each experimental condition have been presented in Fig. 1. Statistical comparisons of the mother and infant data across the four conditions were performed using a two-way analysis of variance with repeated measures. The Newman–Keuls test was employed for individual comparisons between conditions. The mother and infant data were also treated independently in two analyses of variance, comparing the Separation Reunion with the two separation conditions.

RESULTS

Evaluation of the levels of plasma cortisol revealed that mothers and infants responded similarly to the experimental conditions although the overall levels of the infants were significantly higher [$F(1, 6) = 6.05$, $P < 0.05$]. The levels of plasma cortisol varied significantly across the four conditions [$F(3, 18) = 6.92$, $P < 0.003$], both mothers and infants manifesting elevated values following separation (see Fig. 1). The cortisol levels of the separated mother–infant dyads were an average 130 μg/100 ml higher than the Separation Reunion levels ($P < 0.05$). The levels following separation were also an average 106 μg/100 ml over Basal values; however, only the Infant Removal condition was significantly higher because of interanimal variability in Basal values. Although the median Basal values of 184 μg/100 ml for the mothers and 209 μg/100 ml for the infants were both lower than the separation levels, the variance was large due to high basal values in one mother–infant pair.

Comparison of the separation conditions with Separation Reunion data for mothers and infants independently revealed that both mothers [$F(2, 6) = 7.28$, $P < 0.03$] and infants [$F(2, 6) = 7.64$, $P < 0.03$] responded to the 30-min separations with significant elevations of plasma cortisol. The median maternal value following Infant Removal was 94 μg above the median Separation Reunion value, slightly less than the median elevation of 107 μg/100 ml following the mothers' removal from offspring into a novel environment.

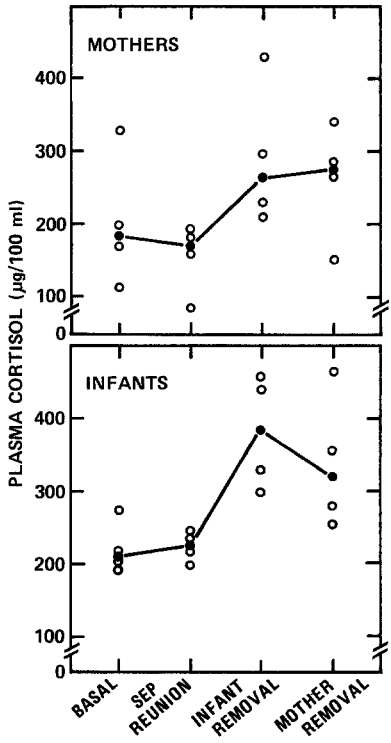


FIG. 1. Median (closed circles) and individual (open circles) levels of plasma cortisol for mother and infant squirrel monkeys.

The infants' levels of plasma cortisol were also significantly elevated following separation. When the infants were removed from the group and housed alone, the median cortisol level was 383 μg/100 ml, an average increase of 157 μg/100 ml over the Separation Reunion levels. The values following Mother Removal also elevated to a median 316 μg/100 ml. These latter elevations occurred despite the fact that three infants were each observed to be on the dorsal surface of the pregnant female at every interval during the 30-min separation. These infants were not observed in the ventral suckling position nor were they emitting distress vocalizations during the separation period. The fourth infant rejected repeated aunting attempts by the pregnant female and also evidenced elevated cortisol levels, although not higher than those of the aunted infants.

DISCUSSION

These results indicate that both mother and infant undergo dramatic elevations in plasma cortisol following involuntary separation. These data are consistent with the prior studies on the behavioral consequences of separation in the squirrel monkey (Jones and Clark, 1973; Kaplan, 1970;

Rosenblum, 1971). This finding also supports our previous work on the adrenal response following separation from the attachment object in individually housed dyads and in surrogate-reared infants (Mendoza *et al.*, in press). However, as demonstrated in the prior experiment, the Separation Reunion condition which involved acute separation followed by immediate reunion did not cause elevations in the mother's and infant's plasma cortisol. The relatively low values obtained in this condition suggest that resumption of contact with the attachment object suppresses or eliminates the reaction to stressful separation.

The variance in Basal values, which probably reflects a variety of unspecified experiential and prior environmental events, did not facilitate comparison with the separation conditions. The hyperresponsiveness of the adrenal in the squirrel monkey which results in inordinately high levels of plasma cortisol (Brown, *et al.*, 1970) also affected the variability and potential range in values. As a consequence the Separation Reunion condition was more appropriate for comparison with the separation data since all animals had received the same handling disturbance. The Separation Reunion values were not significantly different from Basal values but the variance was considerably reduced, presumably because the monkeys had experienced the same synchronized event prior to blood sampling.

A comparison of the two separation conditions indicated that the response to separation was not reduced by the presence of familiar animals. The mothers' levels of plasma cortisol were as elevated following separation in the "home" environment as when they were removed and housed in a novel cage. The infants' adrenal responses were also elevated equivalently in both separation conditions. Moreover, the increases in the infants' plasma cortisol in the Mother Removal condition occurred despite the fact that three infants had been carried by an aunting female. These data indicate that the squirrel monkey infant is markedly agitated while separated and continues to be highly aroused even when aunted by a familiar female during brief separation.

Thus, the interpretation of the cortisol data differs from the conclusion, derived from the observation of disturbance behaviors during sustained separation, that aunting ameliorates the infant's agitation (Rosenblum, 1971). However, extensive procedural differences, such as the infant's age at separation and the length of separation, prevent direct comparison between the two studies. The older infants and longer separation utilized in the Rosenblum study appeared to permit a more affiliative relationship to develop with the aunting female. In contrast, our results indicate that the infant manifests a specific attachment to its mother and does not transfer this affectional tie to an aunting female during the stressful experience of acute separation. Furthermore, the present study suggests that the reduction in the infant's arousal following reunion with its mother is not mediated simply through the dimension of "contact comfort" but rather is due to the reestablishment of the attachment relationship.

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